

A Variational Discrete Crack Element Framework for Dynamic Crack Networks

Nasr M. Ghoniem

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Abstract

We present a Discrete Crack Element (DCE) framework for interacting cracks in two-dimensional isotropic elasticity, designed to preserve crack-tip singular fields through dislocation-based kernels while enabling scalable solution strategies based on constrained variational principles and linear algebra. Cracks are represented as polylines assembled from half-crack elements that carry polarity and local coordinate frames. We formulate static and dynamic governing equations using elastostatic and elastodynamic kernels, compare strong (collocation) and weak (Galerkin / least-squares) enforcement of traction-free conditions, and discuss practical representations of the Burgers distribution. A single-crack example is developed in an 8×8 system using four elements and two-point Gauss enforcement, suitable for hand verification. We then describe polyline crack growth driven by tip criteria, including single-tip kinking, optional remeshing, and time stepping with history-dependent dynamic operators. Finally, we outline pathways to large-scale computations with thousands to millions of cracks using advanced linear algebra, fast summation, and GPU acceleration.

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1 Introduction

Modeling interacting cracks is fundamental to predicting failure in structural materials, geological media, and high-heat-flux components. Classical linear elastic fracture mechanics (LEFM) provides closed-form stress-intensity factors (SIFs) for isolated cracks in idealized domains [1, 2]. Realistic configurations include multiple interacting cracks, curved crack paths, and dynamic loading, requiring numerical methods.

Continuum FEM. Standard displacement-based finite element methods (FEM) approximate fields on a volumetric mesh. Tip singularities are captured only via refinement or special elements; crack growth often requires remeshing unless additional machinery is introduced [3].

Enriched methods (XFEM/GFEM). XFEM introduces discontinuous and near-tip enrichments so cracks need not align with the mesh [4, 5]. These approaches reduce remeshing but add complexity in enrichment management, quadrature, and conditioning.

Boundary integral / BEM. Boundary integral methods reduce dimensionality by using Green’s functions and are naturally suited to crack singularities [6, 7]. Their main computational burden is dense coupling, typically mitigated with fast methods.

Diffuse-interface / phase-field. Phase-field fracture regularizes cracks with a damage field and naturally handles branching and complex topology [8, 9]. However, the representation is volumetric and introduces a regularization length scale.

Dislocation-based and DCE approaches. Dislocation representations of cracks are classical [11, 10]. In DCE approaches, cracks are represented by distributions of dislocations or displacement discontinuities; traction-free conditions and crack-tip singularity are preserved. Traditional dislocation-based DCE often evolves dislocation positions through Peach–Koehler forces, yielding nonlinear equilibrium problems and stiff time integration even for quasi-static analysis.

Summary and objective. This work aims to preserve the physical strengths of dislocation-based crack representations while enabling scalable computation for interacting polyline crack networks. We do so by (i) representing cracks via half-crack elements with polarity; (ii) formulating traction-free conditions in strong and weak forms; (iii) supporting static and dynamic kernels; (iv) enabling growth through local geometry extension; and (v) emphasizing constrained variational and least-squares residual strategies that reduce nonlinear dislocation dynamics to robust linear algebra.

2 Theory

2.1 Polyline cracks as graphs; global and local coordinates

We represent each crack as a connected polyline graph

$$\Gamma = \bigcup_{e=1}^{N_e} \Gamma_e,$$

where each segment Γ_e is straight and adjacent segments meet at junctions.

Let (X_1, X_2) be global coordinates in the plane and X_3 the out-of-plane axis. Each segment (or half-segment element) e has a local orthonormal frame $(\mathbf{e}_1^{(e)}, \mathbf{e}_2^{(e)}, \mathbf{e}_3^{(e)})$:

$$\mathbf{e}_1^{(e)} = \text{unit tangent along the segment}, \quad (1)$$

$$\mathbf{e}_2^{(e)} = \text{unit normal to the crack face}, \quad (2)$$

$$\mathbf{e}_3^{(e)} = \mathbf{e}_1^{(e)} \times \mathbf{e}_2^{(e)}. \quad (3)$$

Local coordinates are $(x_1^{(e)}, x_2^{(e)}, x_3^{(e)})$, with $x_3^{(e)} \equiv X_3$.

Figure 1: polyline crack graph and local frames

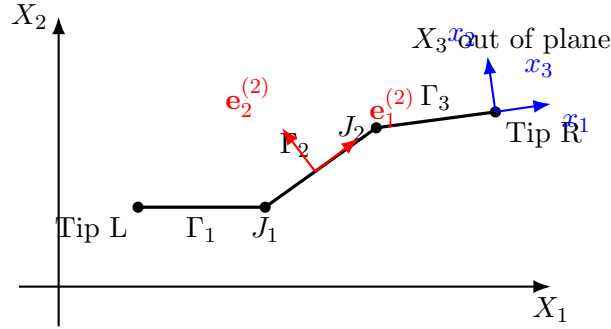


Figure 1: Polyline crack as a graph of straight segments Γ_e connected at junctions J_k . Each segment carries a local frame (x_1, x_2, x_3) .

2.2 Half-crack elements, Burgers vectors, and polarity

We represent crack-face displacement discontinuities using distributions of edge dislocations aligned with local element axes. The Burgers density on element e is

$$\mathbf{b}^{(e)}(\xi) = b_1^{(e)}(\xi) \mathbf{e}_1^{(e)} + b_2^{(e)}(\xi) \mathbf{e}_2^{(e)}, \quad (4)$$

where Mode II corresponds to b_1 and Mode I to b_2 .

Polarity. Define element polarity $\pi^{(e)} \in \{+1, -1\}$ by the right-hand rule:

$$\pi^{(e)} = \begin{cases} +1, & \mathbf{e}_3^{(e)} \parallel +X_3 \quad (x_3 \text{ out of plane}), \\ -1, & \mathbf{e}_3^{(e)} \parallel -X_3 \quad (x_3 \text{ into plane}). \end{cases} \quad (5)$$

Figure 2: half-crack element, local axes, Burgers vector, polarity

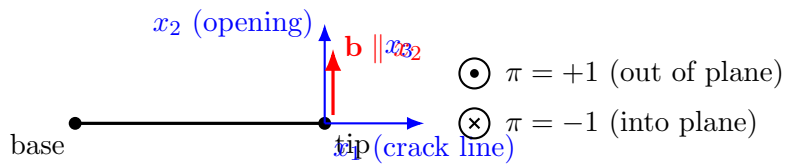


Figure 2: Half-crack element with local axes. Mode-I Burgers vector is aligned with x_2 . Polarity is defined by the right-hand rule on x_3 .

2.3 Field equations; static and dynamic kernels in an infinite medium

We consider plane strain, isotropic elasticity. Let $\boldsymbol{\sigma}(\mathbf{x}, t)$ be stress and $\mathbf{t}(\mathbf{x}, t) = \boldsymbol{\sigma}(\mathbf{x}, t)\mathbf{n}(\mathbf{x})$ traction on crack faces.

Static representation. A dislocation density $\mathbf{b}(\boldsymbol{\xi})$ distributed on Γ produces

$$\sigma_{ij}(\mathbf{x}) = \sigma_{ij}^{\infty}(\mathbf{x}) + \int_{\Gamma} \mathcal{K}_{ijk}^{\text{stat}}(\mathbf{x}, \boldsymbol{\xi}) b_k(\boldsymbol{\xi}) d\Gamma_{\boldsymbol{\xi}}. \quad (6)$$

Dynamic representation. In dynamics (causal wave propagation),

$$\sigma_{ij}(\mathbf{x}, t) = \sigma_{ij}^{\infty}(\mathbf{x}, t) + \int_0^t \int_{\Gamma} \mathcal{K}_{ijk}^{\text{dyn}}(\mathbf{x}, \boldsymbol{\xi}, t - \tau) b_k(\boldsymbol{\xi}, \tau) d\Gamma_{\boldsymbol{\xi}} d\tau. \quad (7)$$

2.4 Traction-free conditions and enforcement: strong vs weak

Traction-free crack faces require

$$\mathbf{t}(\mathbf{x}, t) = \boldsymbol{\sigma}(\mathbf{x}, t)\mathbf{n}(\mathbf{x}) = \mathbf{0}, \quad \mathbf{x} \in \Gamma. \quad (8)$$

2.4.1 Strong form (collocation)

Choose points $\{\mathbf{x}_i\}_{i=1}^{N_t} \subset \Gamma$:

$$\mathbf{t}(\mathbf{x}_i, t) = \mathbf{0}, \quad i = 1, \dots, N_t. \quad (9)$$

2.4.2 Galerkin weak form

For a test space $\mathcal{V}$ on Γ :

$$\int_{\Gamma} \mathbf{v}(\mathbf{x}) \cdot \mathbf{t}(\mathbf{x}, t) d\Gamma = 0, \quad \forall \mathbf{v} \in \mathcal{V}. \quad (10)$$

2.4.3 Least-squares residual (robust discrete weak enforcement)

Let $\mathbf{q}$ denote DOFs representing the Burgers distribution. After discretization, define residual tractions

$$\mathbf{r}(\mathbf{q}) = \mathbf{T}\mathbf{q} + \mathbf{t}^{\infty}.$$

The least-squares formulation solves

$$\min_{\mathbf{q}} \frac{1}{2} \|\mathbf{W}\mathbf{r}(\mathbf{q})\|_2^2 \quad \text{s.t.} \quad \mathbf{C}\mathbf{q} = \mathbf{0}. \quad (11)$$

Physical constraints (closure / gauge). For each connected crack component, enforce a single gauge condition. For Mode I:

$$\int_{\Gamma} b_2(\boldsymbol{\xi}) d\Gamma_{\boldsymbol{\xi}} = 0, \quad (12)$$

equivalently anchoring COD at one reference point and requiring COD closes at the other tip.

2.5 Representations of $b(x)$ and tradeoffs

Along a straight segment parametrized by arc-length s :

1. Discrete dislocations (delta functions):

$$b(s) = \sum_{j=1}^N \beta_j \delta(s - s_j).$$

Pros: natural dislocation physics; explicit singularity. *Cons*: if s_j evolve (EOM-based), equilibrium becomes nonlinear and stiff.

2. Singular-weighted Chebyshev expansion:

$$b(s) = \frac{1}{\sqrt{(a-s)(a+s)}} \sum_{k=0}^p c_k T_k(s/a).$$

Pros: exact tip singularity built in; rapid convergence for smooth remainder. *Cons*: requires careful quadrature and mapping on polylines.

3. Polynomial COD representation (FEM-like):

Approximate COD $w(s)$ piecewise and set $b(s) \propto w'(s)$. *Pros*: standard variational structure and local DOFs. *Cons*: without enrichment/grading, smooths the tip singularity.

3 Constrained Variational Formulation

3.1 Problem Setting

We consider a crack network represented by a collection of parametrized branches. Each branch carries unknown density fields that generate displacement jumps and stress fields. The governing equations are formulated as a constrained variational problem.

3.2 Primal Unknowns

Let the discrete solver unknowns be collected into a vector

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_0 \\ \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_{N_b-1} \end{bmatrix} \in \mathbb{R}^n, \quad (13)$$

where each block $\mathbf{x}_k$ represents the discretized coefficients of the density fields on branch k (e.g. b_I, b_{II} coefficients in a Chebyshev or panel basis).

The local density vector along a branch is

$$\mathbf{q}(s) = b_{II}(s) \mathbf{t}(s) + b_I(s) \mathbf{n}(s), \quad (14)$$

where $\mathbf{t}$ and $\mathbf{n}$ are the local tangent and normal vectors.

3.3 Derived Geometric Jump

The displacement jump (COD/CSD) reconstructed from the solution is

$$\mathbf{J}(s; \mathbf{x}) = \text{CSD}(s; \mathbf{x}) \mathbf{t}(s) + \text{COD}(s; \mathbf{x}) \mathbf{n}(s). \quad (15)$$

This field is *geometric* and represents the physical opening and sliding of the crack.

3.4 Objective Functional

The solver minimizes a functional of the form

$$\min_{\mathbf{x}} f(\mathbf{x}), \quad (16)$$

where f is typically a quadratic functional representing traction residuals, integral equation enforcement, and regularization. Its exact form is not required for the KKT structure.

3.5 Equality Constraints

The problem is subject to equality constraints

$$\mathbf{c}(\mathbf{x}) = \mathbf{0}. \quad (17)$$

These include:

3.6 Junction Compatibility

At a junction vertex v with incident branch ends $\mathcal{I}(v) = \{(p_i, e_i)\}$, the jump must be single-valued:

$$\mathbf{J}_{p_i, e_i}(\mathbf{x}) - \mathbf{J}_{p_0, e_0}(\mathbf{x}) = \mathbf{0}, \quad \forall (p_i, e_i) \in \mathcal{I}(v). \quad (18)$$

3.7 Degree-1 Tip Gauge (Closure)

For a degree-1 vertex v_t , the jump is fixed by a gauge condition, typically chosen as

$$\mathbf{J}(v_t; \mathbf{x}) = \mathbf{0}. \quad (19)$$

This removes the additive integration constant in the jump reconstruction and enforces physical closure at free tips.

3.8 Optional Inequality Constraints

If non-penetration or contact is enforced, inequality constraints are added:

$$\mathbf{g}(\mathbf{x}) \leq \mathbf{0}, \quad (20)$$

for example

$$\text{COD}(s; \mathbf{x}) \geq 0. \quad (21)$$

These are optional and may be inactive in many simulations.

3.9 Lagrangian

The Lagrangian of the constrained problem is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top \mathbf{c}(\mathbf{x}) + \boldsymbol{\mu}^\top \mathbf{g}(\mathbf{x}), \quad (22)$$

where $\boldsymbol{\lambda}$ are multipliers enforcing equality constraints and $\boldsymbol{\mu}$ are multipliers for inequality constraints.

3.10 KKT Conditions

A solution $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ satisfies:

3.11 Stationarity

$$\nabla_{\mathbf{x}} f(\mathbf{x}^*) + \nabla_{\mathbf{x}} \mathbf{c}(\mathbf{x}^*)^\top \boldsymbol{\lambda}^* + \nabla_{\mathbf{x}} \mathbf{g}(\mathbf{x}^*)^\top \boldsymbol{\mu}^* = \mathbf{0}. \quad (23)$$

3.12 Primal Feasibility

$$\mathbf{c}(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{g}(\mathbf{x}^*) \leq \mathbf{0}. \quad (24)$$

3.13 Dual Feasibility

$$\boldsymbol{\mu}^* \geq \mathbf{0}. \quad (25)$$

3.14 Complementary Slackness

$$\boldsymbol{\mu}^* \odot \mathbf{g}(\mathbf{x}^*) = \mathbf{0}. \quad (26)$$

3.15 Linearized KKT System

For quadratic f and linear constraints $\mathbf{c}(\mathbf{x}) = C\mathbf{x} - \mathbf{d}$, the KKT system reduces to the saddle-point problem

$$\begin{bmatrix} H & C^\top \\ C & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{d} \end{bmatrix}. \quad (27)$$

This is the algebraic structure solved by the crack network solver.

3.16 Interpretation of Computed Quantities

- The reconstructed jump $\mathbf{J}$ is a *primal geometric field*.
- The integrated content

$$\mathbf{B} = \int \mathbf{q}(s) \, ds \quad (28)$$

is a kinematic resultant derived from $\mathbf{x}$, not a KKT multiplier.

- True Peach–Koehler or configurational forces arise from shape derivatives of the total energy after solving the KKT system.

3.17 Conclusion

The crack network solver is a constrained variational problem whose governing equations are precisely the Karush–Kuhn–Tucker conditions. Junction compatibility, tip closure, and optional contact are enforced through equality and inequality constraints, while the geometric jump field emerges as a derived primal quantity.

4 Relation Between the Variational Crack Method and the Somigliana Dislocation Formulation

We clarify here the relationship between the classical *Somigliana dislocation formulation* of cracks in linear elasticity and the present *Variational Crack Method (VCM)*. Although the two approaches are mathematically equivalent at the continuous level, they differ fundamentally in formulation, numerical realization, and extensibility to crack networks.

The key distinction is that the Somigliana formulation prescribes or directly solves for a displacement jump density through singular integral equations, whereas VCM treats the jump density as a primary variational unknown obtained by minimizing elastic energy under explicit constraints.

4.1 Somigliana Dislocation Representation

Consider a crack Γ embedded in an infinite, homogeneous, isotropic elastic solid. Let $\mathbf{b}(s)$ denote the displacement jump (dislocation density) along the crack parameterized by arc-length s .

The displacement field at a point $\mathbf{x}$ can be written using Somigliana's identity as

$$u_i(\mathbf{x}) = \int_{\Gamma} U_{ij}(\mathbf{x}, \mathbf{y}) b_j(\mathbf{y}) d\Gamma(\mathbf{y}), \quad (29)$$

where U_{ij} is the displacement Green's function.

The stress field follows as

$$\sigma_{ik}(\mathbf{x}) = \int_{\Gamma} S_{ikj}(\mathbf{x}, \mathbf{y}) b_j(\mathbf{y}) d\Gamma(\mathbf{y}), \quad (30)$$

with S_{ikj} the stress kernel obtained by differentiating U_{ij} .

Enforcing traction-free conditions on the crack faces yields a singular integral equation for $\mathbf{b}(s)$:

$$\int_{\Gamma} \mathbf{K}(s, s') \mathbf{b}(s') ds' = \mathbf{t}^{\text{app}}(s), \quad (31)$$

where $\mathbf{K}$ is a hypersingular traction operator and $\mathbf{t}^{\text{app}}$ is the applied traction from remote loading.

4.2 Variational Crack Method

In the Variational Crack Method, the crack-opening displacement density $\mathbf{b}(s)$ is treated as an unknown field determined by minimizing the total potential energy.

The elastic energy can be written entirely in terms of $\mathbf{b}$:

$$\Pi[\mathbf{b}] = \frac{1}{2} \int_{\Gamma} \int_{\Gamma} \mathbf{b}(s) \cdot \mathbf{K}(s, s') \mathbf{b}(s') ds ds' - \int_{\Gamma} \mathbf{b}(s) \cdot \mathbf{t}^{\text{app}}(s) ds. \quad (32)$$

Additional linear constraints are imposed to enforce:

- compatibility at crack junctions,
- removal of rigid-body (gauge) modes,
- optional anchoring or symmetry conditions.

These constraints are enforced using a Karush–Kuhn–Tucker (KKT) formulation.

4.3 Derivation of the Somigliana Integral Equation from Π

Taking the first variation of the potential energy with respect to $\mathbf{b}$ gives

$$\delta\Pi = \int_{\Gamma} \int_{\Gamma} \delta\mathbf{b}(s) \cdot \mathbf{K}(s, s') \mathbf{b}(s') ds ds' - \int_{\Gamma} \delta\mathbf{b}(s) \cdot \mathbf{t}^{\text{app}}(s) ds. \quad (33)$$

Requiring stationarity for arbitrary admissible variations $\delta\mathbf{b}$ yields

$$\int_{\Gamma} \mathbf{K}(s, s') \mathbf{b}(s') ds' = \mathbf{t}^{\text{app}}(s), \quad (34)$$

which is precisely the classical Somigliana traction integral equation.

Thus, the Somigliana dislocation formulation emerges as the *Euler–Lagrange equation* associated with the VCM energy functional.

4.4 Relation to Classical BEM Kernels

In classical boundary element methods (BEM) for cracks:

- $\mathbf{K}(s, s')$ corresponds to hypersingular traction kernels derived from Green's functions,
- discretization leads to dense, generally non-symmetric collocation systems.

In VCM:

- the same kernels appear inside a quadratic energy form,
- discretization yields symmetric, positive semi-definite stiffness matrices,
- constraints are enforced explicitly through Lagrange multipliers rather than implicitly through collocation.

Hence, each VCM kernel is algebraically equivalent to a BEM influence coefficient, but is used within a variational framework rather than a strong-form integral equation.

4.5 Advantages of the Variational Formulation

The variational formulation offers several key advantages:

- natural treatment of crack junctions and branching networks,
- explicit control of gauge freedom and null modes,
- direct access to configurational (Peach–Koehler) forces via energy derivatives,
- improved robustness for crack propagation and evolving geometries.

4.6 Conclusion

The Variational Crack Method can be interpreted as a constrained, weak, energy-based realization of the classical Somigliana dislocation formulation. While both approaches are mathematically equivalent at the continuous level, the variational framework provides superior flexibility and robustness for crack networks, junctions, and propagation problems.

5 Single Static Crack Representation

We consider a straight crack of length $2a$ lying on the global X_1 axis:

$$\Gamma = \{(x, 0) : x \in [-a, a]\},$$

subjected to remote tension $\sigma_{yy}^\infty = \sigma_0$ and $\sigma_{xy}^\infty = 0$ (pure Mode I).

5.1 Four elements with two DOFs each (8 unknowns total)

Partition $[-a, a]$ into four equal elements:

$$E_1 : [-a, -a/2], \quad E_2 : [-a/2, 0], \quad E_3 : [0, a/2], \quad E_4 : [a/2, a],$$

each of length $h = a/2$, mapped by $x = x_c^{(e)} + \frac{h}{2}\eta$, $\eta \in [-1, 1]$.

To obtain an 8 unknown hand-checkable system, represent the Mode-I Burgers density on each element by a *linear* (two-node) basis multiplied by the square-root tip weight:

$$b_2^{(e)}(x) = w(x) \sum_{m=1}^2 q_{em} N_m(\eta^{(e)}(x)), \quad w(x) = \frac{1}{\sqrt{a^2 - x^2}}, \quad (35)$$

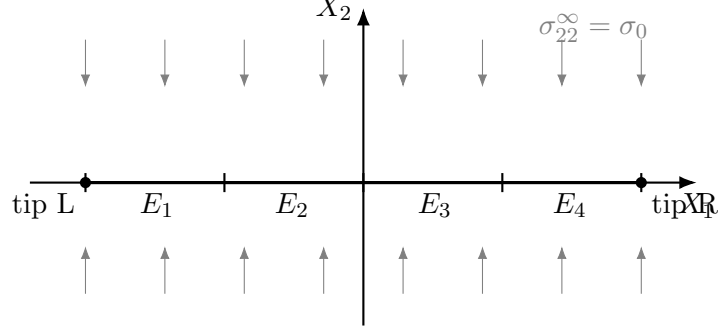


Figure 3: Straight crack on X_1 axis under remote tension, discretized into four quadratic elements (two per half).

with linear shape functions;

$$N_1(\eta) = \frac{1-\eta}{2}, \quad N_2(\eta) = \frac{1+\eta}{2}. \quad (36)$$

Collect the DOFs in the order

$$\mathbf{q} = [q_{11}, q_{12} \mid q_{21}, q_{22} \mid q_{31}, q_{32} \mid q_{41}, q_{42}]^T \in \mathbb{R}^8.$$

5.2 Two Gauss points per element (8 equations total)

Enforce traction-free $\sigma_{yy}(x, 0^+) = 0$ in the two-point Gauss abscissa on each element:

$$\eta_g = \pm \frac{1}{\sqrt{3}}, \quad g = 1, 2, \quad x_{eg} = x_c^{(e)} + \frac{h}{2}\eta_g.$$

The strong-form equations are for each element $e = 1, \dots, 4$ and Gauss point $g = 1, 2$:

$$\sigma_0 + \sum_{e'=1}^4 \int_{E_{e'}} \kappa_{yy}^{\text{stat}}(x_{eg}, 0^+; \xi) b_2^{(e')}(\xi) d\xi = 0. \quad (37)$$

Substituting (35) produces the linear system

$$\sum_{e'=1}^4 \sum_{m=1}^2 A_{(e,g),(e',m)} q_{e'm} = -\sigma_0, \quad e = 1..4, \quad g = 1, 2, \quad (38)$$

where

$$A_{(e,g),(e',m)} = \int_{E_{e'}} \kappa_{yy}^{\text{stat}}(x_{eg}, 0^+; \xi) w(\xi) N_m(\eta^{(e')}(\xi)) d\xi. \quad (39)$$

Explicit quadrature form (2-point Gauss on the source element). Approximate (39) using the two-point Gauss quadrature on the source element $E_{e'}$:

$$A_{(e,g),(e',m)} \approx \sum_{k=1}^2 \kappa_{yy}^{\text{stat}}(x_{eg}, 0^+; \xi_{e'k}) w(\xi_{e'k}) N_m(\eta_k) \underbrace{\left(\frac{h}{2}\right)}_J \underbrace{w_k}_{=1}, \quad \eta_k = \pm \frac{1}{\sqrt{3}}, \quad (40)$$

with Gauss point source $\xi_{e'k} = x_c^{(e')} + \frac{h}{2}\eta_k$.

5.3 Example: explicit form of A_{12}

Using the global row/column ordering

$$\text{rows: } r = 2(e - 1) + g \in \{1, \dots, 8\}, \quad \text{cols: } c = 2(e' - 1) + m \in \{1, \dots, 8\},$$

the entry A_{12} corresponds to $(e, g) = (1, 1)$ and $(e', m) = (1, 2)$, i.e. the first Gauss equation on element 1 and the second DOF on element 1:

$$\begin{aligned} A_{12} &= \int_{E_1} \kappa_{yy}^{\text{stat}}(x_{11}, 0^+; \xi) w(\xi) N_2(\eta^{(1)}(\xi)) d\xi \\ &\approx \sum_{k=1}^2 \kappa_{yy}^{\text{stat}}(x_{11}, 0^+; \xi_{1k}) w(\xi_{1k}) N_2(\eta_k) \left(\frac{h}{2}\right), \quad \eta_k = \pm \frac{1}{\sqrt{3}}, \quad \xi_{1k} = x_c^{(1)} + \frac{h}{2} \eta_k, \end{aligned} \quad (41)$$

with $N_2(\eta_k) = \frac{1+\eta_k}{2}$ and $x_{11} = x_c^{(1)} + \frac{h}{2} \left(-\frac{1}{\sqrt{3}}\right)$.

5.4 Assembled 8×8 block matrix structure

With 4 elements and 2 DOFs per element, the system (38) assembles as

$$\mathbf{A} \mathbf{q} = \mathbf{f}, \quad \mathbf{f} = -\sigma_0 \mathbf{1}_8, \quad (42)$$

where $\mathbf{A} \in \mathbb{R}^{8 \times 8}$ has a 4×4 block structure with 2×2 blocks:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}^{11} & \mathbf{A}^{12} & \mathbf{A}^{13} & \mathbf{A}^{14} \\ \mathbf{A}^{21} & \mathbf{A}^{22} & \mathbf{A}^{23} & \mathbf{A}^{24} \\ \mathbf{A}^{31} & \mathbf{A}^{32} & \mathbf{A}^{33} & \mathbf{A}^{34} \\ \mathbf{A}^{41} & \mathbf{A}^{42} & \mathbf{A}^{43} & \mathbf{A}^{44} \end{bmatrix}, \quad \mathbf{A}^{ee'} \in \mathbb{R}^{2 \times 2}, \quad (43)$$

and each block is

$$\mathbf{A}^{ee'} = \begin{bmatrix} A_{(e,1),(e',1)} & A_{(e,1),(e',2)} \\ A_{(e,2),(e',1)} & A_{(e,2),(e',2)} \end{bmatrix}. \quad (44)$$

5.5 Closure constraint and constrained solution

Impose one Mode-I closure (gauge) constraint for the connected crack:

$$\int_{-a}^a b_2(x) dx = 0 \quad \implies \quad \mathbf{C} \mathbf{q} = 0, \quad (45)$$

where

$$(\mathbf{C})_c = \int_{E_{e(c)}} w(x) N_{m(c)}(\eta^{(e(c))}(x)) dx,$$

with $e(c)$, $m(c)$ mapping global DOF index c to element and local basis index.

A convenient constrained solve is the KKT form:

$$\begin{bmatrix} \mathbf{A} & \mathbf{C}^T \\ \mathbf{C} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}, \quad (46)$$

or the least-squares analog $\min_{\mathbf{q}} \frac{1}{2} \|\mathbf{A} \mathbf{q} - \mathbf{f}\|_2^2$ subject to $\mathbf{C} \mathbf{q} = 0$.

5.6 Displacement jump (kinematics)

For a crack (or interface) Γ , define the *displacement jump*:

$$\boxed{\mathbf{J}(\mathbf{x}) \equiv \mathbf{u}^+(\mathbf{x}) - \mathbf{u}^-(\mathbf{x}), \quad \mathbf{x} \in \Gamma.} \quad (47)$$

This is a *purely kinematic* quantity: it measures separation of the two crack faces. It does *not* represent force or stress.

Tip and junction expectations. At a free crack tip (degree-1 vertex),

$$\boxed{\lim_{r \rightarrow 0} \mathbf{J}(r) = \mathbf{0}.} \quad (48)$$

At a junction (degree ≥ 2 vertex),

$$\boxed{\mathbf{J}_{\text{branch } 1}(v) = \mathbf{J}_{\text{branch } 2}(v) = \dots = \mathbf{J}(v).} \quad (49)$$

—

5.7 COD and CSD: local components of the jump

Let $\mathbf{t}$ be the unit tangent and $\mathbf{n}$ the unit normal to the crack midline. Define:

$$\boxed{\text{CSD}(s) = \mathbf{J}(s) \cdot \mathbf{t}(s), \quad \text{COD}(s) = \mathbf{J}(s) \cdot \mathbf{n}(s).} \quad (50)$$

Equivalently,

$$\boxed{\mathbf{J}(s) = \text{CSD}(s) \mathbf{t}(s) + \text{COD}(s) \mathbf{n}(s).} \quad (51)$$

COD and CSD are simply the *components* of the jump vector in the local crack frame.

—

5.8 Dislocation density and Burgers content

In dislocation-density formulations, the jump gradient along arc length s is written as

$$\boxed{\frac{\partial \mathbf{J}}{\partial s}(s) = -(b_{II}(s) \mathbf{t}(s) + b_I(s) \mathbf{n}(s)).} \quad (52)$$

The *Burgers content* of a branch is the integral

$$\boxed{\mathbf{B}_{\text{branch}} = \int_{\text{branch}} (b_{II} \mathbf{t} + b_I \mathbf{n}) ds.} \quad (53)$$

This quantity:

- is *topological/kinematic*, not a force,
- need not vanish at a free tip,
- must satisfy neutrality at junctions:

$$\boxed{\sum_{\text{branches at } v} \mathbf{B}_{\text{branch}} = \mathbf{0}.} \quad (54)$$

—

5.9 How stresses arise

The stress field is generated by elastic kernels acting on the *density* $\partial_s \mathbf{J}$:

$$\boxed{\boldsymbol{\sigma}(\mathbf{x}) = \boldsymbol{\sigma}^\infty(\mathbf{x}) + \int_{\Gamma} \mathcal{K}(\mathbf{x}, \mathbf{x}') \frac{\partial \mathbf{J}}{\partial s}(\mathbf{x}') ds'.} \quad (55)$$

Key implication. An additive affine change in $\mathbf{J}(s)$ does *not* alter the stress field, because stresses depend on gradients of $\mathbf{J}$, not its absolute value.

5.10 TikZ figures

5.10.1 Jump, COD, and CSD on a crack

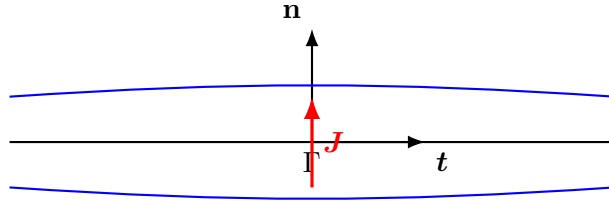


Figure 4: Displacement jump $\mathbf{J} = \boldsymbol{\mathfrak{u}}^+ - \boldsymbol{\mathfrak{u}}^-$ and its local components COD and CSD along a crack.

5.10.2 Jump gradient, Burgers content, and endpoint relation

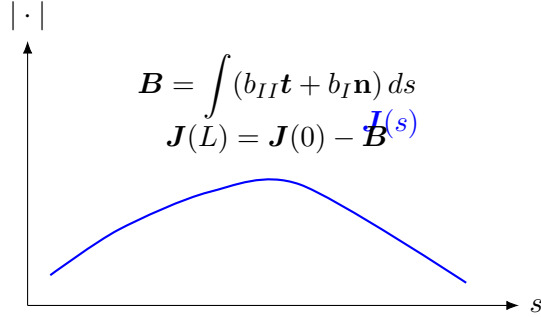


Figure 5: The Burgers content $\mathbf{B}$ is the integral of the jump gradient and relates endpoint values of $\mathbf{J}$.

5.11 Summary

- $\mathbf{J}$ is a displacement discontinuity (kinematic).
- COD/CSD are its local components.
- $\partial_s \mathbf{J}$ produces stress via elastic kernels.
- $\mathbf{B}$ is integrated dislocation content (topological).
- Tip closure errors in reconstruction do not imply solver errors.

6 The Static Traction Kernel

6.1 Single Crack Kernel

For Mode I we take edge dislocations whose Burgers vectors are aligned with the crack normal (local x_2), i.e. $\mathbf{b} = b_2 \mathbf{e}_2$. In an infinitely isotropic medium (plane deformation), the contribution of a single edge dislocation located at ξ on the x -axis to the stress component σ_{yy} at a field point (x, y) is [11, 10]

$$\sigma_{yy}(x, y; \xi) = \kappa_{yy}^{\text{stat}}(x, y; \xi) b_2, \quad \kappa_{yy}^{\text{stat}}(x, y; \xi) = -\frac{\mu}{2\pi(1-\nu)} \frac{(x-\xi) ((x-\xi)^2 + 3y^2)}{((x-\xi)^2 + y^2)^2}, \quad (56)$$

where μ is shear modulus and ν Poisson ratio. On the crack line (approaching from one side), $y \rightarrow 0^+$, this reduces (in the sense of Cauchy principal value) to the singular kernel

$$\kappa_{yy}^{\text{stat}}(x, 0^+; \xi) = -\frac{\mu}{2\pi(1-\nu)} \text{p.v.} \left(\frac{1}{x-\xi} \right). \quad (57)$$

6.2 Interacting Cracks



Figure 6: Two interacting cracks with independent local coordinate systems.

We consider two distinct cracks $\Gamma^{(a)}$ (target) and $\Gamma^{(b)}$ (source), arbitrarily oriented in the global (X_1, X_2) plane. Let the field point be

$$\mathbf{X} = (X_1, X_2) \in \Gamma^{(a)}, \quad \mathbf{\Xi} = (\Xi_1, \Xi_2) \in \Gamma^{(b)}.$$

Define

$$\Delta X = X_1 - \Xi_1, \quad \Delta Y = X_2 - \Xi_2, \quad R^2 = \Delta X^2 + \Delta Y^2.$$

All kernels below are written in the form

$$\sigma_{ij} = \mathcal{M} \tilde{\sigma}_{ij},$$

It is convenient to write the isotropic Volterra edge-dislocation stresses as

$$\sigma_{ij}(\mathbf{X}) = \mathcal{M} \tilde{\sigma}_{ij}(\Delta X, \Delta Y),$$

where the elastic multiplier $\mathcal{M}$ is expressed in terms of the Kolosov constant

$$\kappa = \begin{cases} 3 - 4\nu, & \text{plane strain,} \\ \frac{3 - \nu}{1 + \nu}, & \text{plane stress.} \end{cases}$$

With this convention,

$$\mathcal{M} = \frac{2\mu}{\pi(\kappa + 1)}$$

so that

$$\mathcal{M} = \begin{cases} \frac{\mu}{2\pi(1-\nu)}, & \text{plane strain,} \\ \frac{\mu(1+\nu)}{2\pi}, & \text{plane stress.} \end{cases}$$

(Equivalently, using $\mu = E/[2(1+\nu)]$, one has $\mathcal{M}_{\text{ps}} = E/(4\pi)$.)

Global stress kernel for an edge dislocation

Let an edge dislocation be located at Ξ with line direction parallel to X_3 and Burgers vector

$$\mathbf{b} = (b_1, b_2)$$

expressed in the *global* frame. The resulting stress field in an infinite medium is

$$\sigma_{ij}(\mathbf{X}) = \sum_{\alpha=1}^2 \kappa_{ij,\alpha}^{\text{stat}}(\mathbf{X} - \Xi) b_{\alpha}, \quad (58)$$

with explicit components

$$\begin{aligned} \sigma_{11} &= -\mathcal{M} \left[b_1 \frac{\Delta Y (3\Delta X^2 + \Delta Y^2)}{R^4} - b_2 \frac{\Delta X (\Delta X^2 - \Delta Y^2)}{R^4} \right], \\ \sigma_{22} &= \mathcal{M} \left[b_1 \frac{\Delta Y (\Delta X^2 - \Delta Y^2)}{R^4} - b_2 \frac{\Delta X (\Delta X^2 + 3\Delta Y^2)}{R^4} \right], \\ \sigma_{12} &= \mathcal{M} \left[b_1 \frac{\Delta X (\Delta X^2 - \Delta Y^2)}{R^4} + b_2 \frac{\Delta Y (\Delta X^2 - \Delta Y^2)}{R^4} \right]. \end{aligned} \quad (59)$$

The kernel $\kappa_{ij,\alpha}^{\text{stat}}$ is *invariant*; all orientation and polarity information enters through $\mathbf{b}$.

6.2.1 Local Burgers components, polarity, and global mapping

Let crack $\Gamma^{(b)}$ have local unit tangent $\mathbf{e}_1^{(b)}$, unit normal $\mathbf{e}_2^{(b)}$, and polarity $\pi^{(b)} = \pm 1$. Mode-I and Mode-II Burgers densities are defined locally as

$$b_I^{(b)}(\xi) \mathbf{e}_2^{(b)}, \quad b_{II}^{(b)}(\xi) \mathbf{e}_1^{(b)}.$$

The *global* Burgers vector entering (59) is

$$\mathbf{b}_{\text{global}}^{(b)}(\Xi) = \pi^{(b)} \left(b_{II}^{(b)}(\Xi) \mathbf{e}_1^{(b)} + b_I^{(b)}(\Xi) \mathbf{e}_2^{(b)} \right). \quad (60)$$

If the source crack is oriented at angle θ_b with respect to X_1 ,

$$\mathbf{e}_1^{(b)} = \begin{bmatrix} \cos \theta_b \\ \sin \theta_b \end{bmatrix}, \quad \mathbf{e}_2^{(b)} = \begin{bmatrix} -\sin \theta_b \\ \cos \theta_b \end{bmatrix},$$

then

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \pi^{(b)} \begin{bmatrix} \cos \theta_b & -\sin \theta_b \\ \sin \theta_b & \cos \theta_b \end{bmatrix} \begin{bmatrix} b_{II}^{(b)} \\ b_I^{(b)} \end{bmatrix}. \quad (61)$$

6.2.2 Traction projection on the target crack

Let crack $\Gamma^{(a)}$ have unit tangent $\mathbf{e}_1^{(a)}$ and unit normal $\mathbf{e}_2^{(a)}$. The global traction on $\Gamma^{(a)}$ is

$$\mathbf{t}_{\text{global}}(\mathbf{X}) = \boldsymbol{\sigma}(\mathbf{X}) \mathbf{e}_2^{(a)}.$$

Normal and shear tractions in the *local* frame of crack a are

$$t_n^{(a)} = \mathbf{e}_2^{(a)} \cdot \boldsymbol{\sigma} \mathbf{e}_2^{(a)}, \quad t_s^{(a)} = \mathbf{e}_1^{(a)} \cdot \boldsymbol{\sigma} \mathbf{e}_2^{(a)}. \quad (62)$$

6.2.3 Explicit two-crack interaction operator

Combining (59)–(62), the interaction from crack b to crack a is

$$\begin{bmatrix} t_s^{(a)} \\ t_n^{(a)} \end{bmatrix}(\mathbf{X}) = \int_{\Gamma^{(b)}} \mathbf{K}^{ab}(\mathbf{X}, \Xi) \begin{bmatrix} b_{II}^{(b)}(\Xi) \\ b_I^{(b)}(\Xi) \end{bmatrix} d\Gamma_\Xi, \quad (63)$$

where the interaction kernel is

$$\mathbf{K}^{ab} = \pi^{(b)} \begin{bmatrix} \mathbf{e}_1^{(a)} \cdot \boldsymbol{\kappa}_{:,1} \mathbf{e}_2^{(a)} & \mathbf{e}_1^{(a)} \cdot \boldsymbol{\kappa}_{:,2} \mathbf{e}_2^{(a)} \\ \mathbf{e}_2^{(a)} \cdot \boldsymbol{\kappa}_{:,1} \mathbf{e}_2^{(a)} & \mathbf{e}_2^{(a)} \cdot \boldsymbol{\kappa}_{:,2} \mathbf{e}_2^{(a)} \end{bmatrix}, \quad (64)$$

with $\boldsymbol{\kappa}_{ij,\alpha}$ given explicitly by (59).

The kernel is universal; crack orientation and polarity enter only through the rotation (61) and scalar multiplier $\pi^{(b)}$.

7 Mixed-Mode Formulation

Mixed-mode loading is represented by simultaneous Mode I and Mode II Burgers densities:

$$\begin{bmatrix} t_n \\ t_s \end{bmatrix} = \begin{bmatrix} \kappa_{nn}^{(I)} & \kappa_{nn}^{(II)} \\ \kappa_{ss}^{(I)} & \kappa_{ss}^{(II)} \end{bmatrix} \begin{bmatrix} b_I \\ b_{II} \end{bmatrix}.$$

The discrete system becomes

$$\begin{bmatrix} \mathbf{A}_{II} & \mathbf{A}_{II} \\ \mathbf{A}_{II}^\top & \mathbf{A}_{III} \end{bmatrix} \begin{bmatrix} \mathbf{q}_I \\ \mathbf{q}_{II} \end{bmatrix} + \begin{bmatrix} \mathbf{f}_I \\ \mathbf{f}_{II} \end{bmatrix} = \mathbf{0},$$

with closure constraints applied independently to Mode I and Mode II.

Mixed-mode Kink Criterion

At a crack tip,

$$\theta_c = 2 \arctan \left[\frac{1}{4} \left(\frac{K_I}{K_{II}} - \sqrt{\left(\frac{K_I}{K_{II}} \right)^2 + 8} \right) \right],$$

(Maximum hoop stress criterion). Only the active tip is extended.

8 Evaluation of Peach–Koehler Force and Stress Intensity Factors in the Variational DCE Formulation

In the present variational discrete crack element (DCE) formulation, the crack is represented by distributed Burgers densities that satisfy traction-free conditions in a weak (variational) sense. Unlike classical dislocation dynamics approaches, individual dislocations are not evolved via equations of motion. Nevertheless, the Peach–Koehler (PK) force and the associated stress intensity factors can be extracted in a consistent and rigorous manner from the converged variational solution.

8.1 Crack-tip representation in the variational setting

Consider a crack tip located at $\mathbf{X}_{\text{tip}}$ associated with a terminal half-crack element. Within the variational formulation, the Burgers density near the tip is represented as

$$b_\alpha(\xi) = \sum_{j=1}^{n_e} N_j(\xi) q_{j\alpha}, \quad \alpha \in \{I, II\}, \quad (65)$$

where N_j are shape functions defined on the half-crack element and $q_{j\alpha}$ are the solved degrees of freedom.

The crack-tip singularity is not imposed explicitly but emerges naturally from the kernel and the admissible function space. Consequently, the crack-tip force and energy release rate must be evaluated as *derived quantities*.

8.2 Variational definition of the Peach–Koehler force

In the absence of explicit dislocation motion, the Peach–Koehler force is defined energetically as the configurational force associated with a virtual advance of the crack tip.

Consider a virtual extension δa of the crack tip in the local tangential direction $\mathbf{e}_1$. The corresponding variation of the total potential energy Π is

$$\delta \Pi = -f_1^{\text{tip}} \delta a, \quad (66)$$

which defines the Peach–Koehler force component

$$\boxed{f_1^{\text{tip}} = -\frac{\partial \Pi}{\partial a}}. \quad (67)$$

Within the variational DCE formulation, Π consists solely of elastic energy and work of external tractions. Since the solution satisfies the Euler–Lagrange equations, the derivative (67) may be evaluated by differentiating only the geometry-dependent terms of the functional.

8.3 Equivalent traction-based expression

An equivalent and numerically convenient expression for the crack-tip PK force is obtained by projecting the stress field generated by the variational solution onto the crack-tip normal:

$$\boxed{f_1^{\text{tip}} = \lim_{\epsilon \rightarrow 0^+} \int_{-\epsilon}^{\epsilon} \sigma_{22}(x_1, 0) dx_1}, \quad (68)$$

where $(x_1, 0)$ denotes coordinates in the local crack-tip frame.

The stress σ_{22} is evaluated using the same global kernels and solved Burgers densities as used in the assembly of the variational system. No additional assumptions regarding dislocation motion are required.

8.4 Relation to energy release rate

By configurational force theory, the energy release rate is given by

$$\boxed{G = f_1^{\text{tip}}}. \quad (69)$$

This equality holds independently of whether the crack is modeled via discrete dislocations or via a variational Burgers density field, provided that the underlying elastic fields are identical.

8.5 Extraction of stress intensity factors

Once the energy release rate is known, the stress intensity factors follow from standard linear elastic fracture mechanics relations.

For mixed-mode loading,

$$G = \begin{cases} \frac{1-\nu^2}{E} (K_I^2 + K_{II}^2), & \text{plane stress,} \\ \frac{1-\nu}{2\mu} (K_I^2 + K_{II}^2), & \text{plane strain.} \end{cases} \quad (70)$$

In cases where the crack-tip mode mixity is known (e.g. by symmetry or by a local symmetry condition), individual stress intensity factors may be obtained directly. For example, under pure Mode I conditions in plane strain,

$$K_I = \sqrt{\frac{2\mu}{1-\nu} f_1^{\text{tip}}}. \quad (71)$$

Alternatively, both K_I and K_{II} may be extracted by fitting the near-tip stress field obtained from the variational solution to the asymptotic LEFM form,

$$\sigma_{ij}(r, \theta) = \frac{1}{\sqrt{2\pi r}} \left[K_I \Sigma_{ij}^{(I)}(\theta) + K_{II} \Sigma_{ij}^{(II)}(\theta) \right] + O(\sqrt{r}), \quad (72)$$

where $\Sigma_{ij}^{(I)}$ and $\Sigma_{ij}^{(II)}$ are the universal angular functions.

8.6 Remarks on consistency and advantages

The present procedure:

- avoids explicit dislocation equations of motion,
- relies solely on quantities already available in the variational solution,
- preserves the correct crack-tip singular fields through the elastic kernel,
- and is directly extensible to dynamic fracture by replacing static kernels with retarded kernels.

Thus, the Peach–Koehler force and stress intensity factors emerge naturally as post-processed quantities within the variational DCE framework, maintaining full consistency with classical fracture mechanics while enabling large-scale interacting crack simulations.

8.7 Computation of the Crack-Tip Peach–Koehler Force

This subsection describes how to compute the crack-tip Peach–Koehler (PK) force from the converged variational DCE solution, *without* invoking dislocation dynamics. Two equivalent routes are available: (i) an energetic (configurational) derivative of the total potential energy Π , and (ii) a traction-based line integral local to the tip. In practice, the traction-based route is often more robust numerically because it avoids differentiating dense geometry-dependent operators.

8.7.1 Total potential energy functional Π

Let the unknowns be the vector of Burgers degrees of freedom (DOF) $\mathbf{q}$ that parameterize the Burgers density on all half-crack elements:

$$\mathbf{b}(\xi) = \pi^{(e)} \sum_{j \in e} N_j(\xi) q_j,$$

with $\pi^{(e)} = \pm 1$ the element polarity. The (static) variational formulation may be written in the standard quadratic form

$$\Pi(\mathbf{q}; a) = \frac{1}{2} \mathbf{q}^\top \mathbf{A}(a) \mathbf{q} + \mathbf{q}^\top \mathbf{f}(a) + \Pi_0(a), \quad (73)$$

where a denotes a crack-length (or, more generally, a crack-tip coordinate) parameter controlling the geometry. Here:

- $\mathbf{A}(a)$ is the symmetric positive-semidefinite influence operator obtained from integrating the dislocation traction kernels against the trial/test spaces (e.g. Galerkin or least-squares normal equations);
- $\mathbf{f}(a)$ is the load vector induced by remote/applied tractions/stresses (and any prescribed tractions on crack faces);
- $\Pi_0(a)$ collects terms independent of $\mathbf{q}$ (e.g. far-field energy reference) and does not affect the stationary condition in $\mathbf{q}$.

The converged variational solution satisfies the Euler–Lagrange (KKT) system

$$\nabla_{\mathbf{q}} \Pi(\mathbf{q}; a) + \mathbf{C}^\top \boldsymbol{\lambda} = \mathbf{0}, \quad \mathbf{C} \mathbf{q} = \mathbf{0},$$

where $\mathbf{C}$ enforces one closure/gauge constraint per connected crack component.

8.7.2 Configurational (energetic) definition of the PK force

The crack-tip PK force component in the local tangential direction is defined by

$$f_1^{\text{tip}} = - \frac{d}{da} \Pi(\mathbf{q}^*(a); a), \quad (74)$$

where $\mathbf{q}^*(a)$ denotes the stationary solution at geometry parameter a .

Because $\mathbf{q}^*(a)$ is stationary with respect to admissible variations (enforcing the constraints), the total derivative reduces to the *partial* derivative with respect to geometry (envelope theorem / stationarity):

$$f_1^{\text{tip}} = - \left. \frac{\partial \Pi(\mathbf{q}; a)}{\partial a} \right|_{\mathbf{q}=\mathbf{q}^*(a)}. \quad (75)$$

Substituting (73) gives

$$f_1^{\text{tip}} = - \frac{1}{2} \mathbf{q}^{*\top} \frac{\partial \mathbf{A}}{\partial a} \mathbf{q}^* - \mathbf{q}^{*\top} \frac{\partial \mathbf{f}}{\partial a} - \frac{\partial \Pi_0}{\partial a}. \quad (76)$$

Practical evaluation of $\partial \mathbf{A}/\partial a$ and $\partial \mathbf{f}/\partial a$. In the present setting, $\mathbf{A}$ and $\mathbf{f}$ depend on geometry through source/field coordinates in the kernels (and through Jacobians of the element mappings). Analytically differentiating dense kernel integrals is possible but cumbersome; a robust practical method is a *shape derivative by finite difference*:

$$f_1^{\text{tip}} \approx -\frac{\Pi(\mathbf{q}^*(a); a + \Delta a) - \Pi(\mathbf{q}^*(a); a - \Delta a)}{2\Delta a}, \quad (77)$$

where $\Pi(\mathbf{q}^*(a); \cdot)$ is evaluated by reassembling $\mathbf{A}$ and $\mathbf{f}$ at the perturbed geometry while holding $\mathbf{q}^*(a)$ fixed (consistent with (75)).

Expected accuracy. If (77) is used with a centered difference and sufficiently smooth geometry perturbations, the truncation error is $O(\Delta a^2)$. However, the overall accuracy is limited by: (i) quadrature error in assembling $\mathbf{A}$ and $\mathbf{f}$, (ii) discretization error in the near-tip representation of $b(x)$, and (iii) conditioning of the constrained system. In practice, Δa should be chosen comparable to a small fraction of the smallest element length near the tip (e.g. 10^{-2} – 10^{-1} of that length) and verified by step refinement.

Comment. Although (76) is formally exact, evaluating it accurately at large scale requires either analytic kernel derivatives or repeated operator assembly. This is why the traction-based method below is frequently preferred.

8.7.3 Traction-based computation (preferred in practice)

An equivalent and often simpler estimator is

$$f_1^{\text{tip}} = \lim_{\epsilon \rightarrow 0^+} \int_{-\epsilon}^{\epsilon} \sigma_{22}(x_1, 0^+) dx_1, \quad (78)$$

where $(x_1, 0)$ are local coordinates centered at the crack tip and 0^+ indicates evaluation on one crack face (a consistent side must be chosen).

Is (78) easier than the energy derivative? Yes, typically:

- it requires no differentiation of geometry-dependent dense operators,
- it uses the same stress evaluation routines already needed for post-processing,
- it can be computed locally near the tip and is therefore compatible with large crack networks.

How to evaluate the limit. In computations, (78) is replaced by an integral over a small but finite window $[-\epsilon, \epsilon]$:

$$f_1^{\text{tip}}(\epsilon) = \int_{-\epsilon}^{\epsilon} \sigma_{22}(x_1, 0^+) dx_1, \quad f_1^{\text{tip}} = \lim_{\epsilon \rightarrow 0^+} f_1^{\text{tip}}(\epsilon). \quad (79)$$

A practical procedure is:

1. Choose a sequence ϵ_m decreasing geometrically (e.g. $\epsilon_{m+1} = \epsilon_m/2$) with ϵ_1 on the order of the near-tip element length.
2. For each ϵ_m , evaluate $\sigma_{22}(x_1, 0^+)$ at quadrature points $x_{1k} \in [-\epsilon_m, \epsilon_m]$ and compute

$$f_1^{\text{tip}}(\epsilon_m) \approx \sum_k w_k \sigma_{22}(x_{1k}, 0^+).$$

3. Extrapolate to $\epsilon \rightarrow 0^+$ (e.g. Richardson extrapolation) or take the stabilized plateau value if $f_1^{\text{tip}}(\epsilon_m)$ converges.

Quadrature choice and singularity handling. Because σ_{22} is singular at the tip, direct sampling at $x_1 = 0$ must be avoided. Use quadrature that excludes the singular point (Gauss points do) and take ϵ not smaller than the smallest resolvable length scale of the discretization. A robust choice is to integrate on each side separately,

$$f_1^{\text{tip}}(\epsilon) = \int_{-\epsilon}^0 \sigma_{22}(x_1, 0^+) dx_1 + \int_0^\epsilon \sigma_{22}(x_1, 0^+) dx_1,$$

with symmetric quadrature rules on $[-\epsilon, 0]$ and $[0, \epsilon]$.

Expected accuracy. If $\sigma_{22}(x_1, 0^+)$ is evaluated accurately and the near-tip discretization captures the correct asymptotic behavior, the ϵ -window integral converges to a stable value. The observed convergence rate is typically governed by the near-tip representation of $b(x)$ and the stress evaluation quadrature; with singular-weighted basis functions or Chebyshev expansions, convergence is substantially improved relative to plain polynomial bases.

8.7.4 From PK force to K and G

Once f_1^{tip} is obtained by either method above, the energy release rate follows from

$$\boxed{G = f_1^{\text{tip}}.} \quad (80)$$

Stress intensity factors then follow from standard LEFM relations. For pure Mode I in plane strain,

$$\boxed{K_I = \sqrt{\frac{2\mu}{1-\nu}} f_1^{\text{tip}}.} \quad (81)$$

For mixed mode,

$$G = \begin{cases} \frac{1-\nu^2}{E} (K_I^2 + K_{II}^2), & \text{plane stress,} \\ \frac{1-\nu}{2\mu} (K_I^2 + K_{II}^2), & \text{plane strain,} \end{cases}$$

and (K_I, K_{II}) may be recovered either by near-tip field fitting or by an additional mode-mixity condition (e.g. local symmetry at the tip).

9 Crack Growth Criteria and Propagation Algorithms

Crack propagation within the discrete crack element (DCE) framework is treated as the extension of an existing half-crack element by the nucleation of an additional half-crack at a crack tip. The direction, driving force, and admissibility of such growth are governed by fracture criteria derived from continuum fracture mechanics, augmented when needed by stochastic or microstructural considerations.

In this section, we summarize (i) criteria for determining the crack growth direction, (ii) methods for computing stress intensity factors and energy release rates within the present formulation, and (iii) propagation criteria based on deterministic thresholds or stochastic activation.

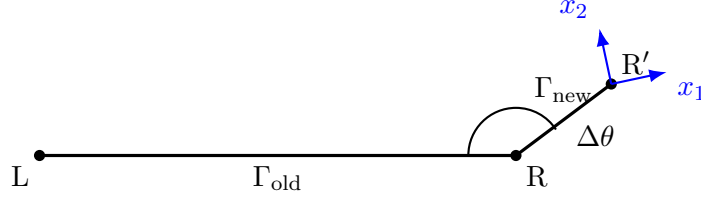


Figure 7: Single-tip extension: append a new segment at the active (right) tip. Old DOFs remain; new DOFs are added only on the appended segment.

9.1 Crack growth direction

Consider a crack tip located at $\mathbf{X}_{\text{tip}}$ with local coordinate system (x_1, x_2) aligned with the current crack segment. A candidate extension is introduced as a new half-crack making an angle θ with respect to the local x_1 axis.

Several classical criteria are commonly employed to determine θ :

Maximum hoop stress criterion (MTS). The crack propagates in the direction that maximizes the circumferential (hoop) stress $\sigma_{\theta\theta}$ ahead of the tip. For mixed-mode loading (K_I, K_{II}) , this yields the condition

$$\frac{\partial \sigma_{\theta\theta}(r, \theta)}{\partial \theta} = 0, \quad (82)$$

which leads to the well-known analytical solution

$$\theta_{\text{MTS}} = 2 \arctan \left[\frac{1}{4} \left(\frac{K_I}{K_{II}} \pm \sqrt{\left(\frac{K_I}{K_{II}} \right)^2 + 8} \right) \right]. \quad (83)$$

This criterion is widely used due to its simplicity and robustness.

Minimum strain energy density criterion (SED). The crack grows in the direction that minimizes the strain energy density factor $S(\theta)$ near the tip. The preferred direction satisfies

$$\frac{\partial S(\theta)}{\partial \theta} = 0, \quad (84)$$

where $S(\theta)$ is expressed in terms of K_I , K_{II} , and the elastic constants. This criterion is particularly attractive when both modes contribute significantly.

Local symmetry criterion. Crack growth is assumed to occur in a direction such that the shear mode vanishes at the new tip:

$$K_{II}(\theta) = 0. \quad (85)$$

This condition enforces symmetry of the near-tip stress field and is naturally compatible with dislocation-based formulations.

Geometric and obstacle-aware criteria. In heterogeneous materials, the growth direction may be constrained to avoid obstacles such as inclusions, grain boundaries, or pre-existing defects. In such cases, θ is chosen to minimize a cost functional incorporating both fracture driving force and geometric constraints.

9.2 Evaluation of stress intensity factors and energy release rate

Within the DCE framework, stress intensity factors (SIFs) and energy release rates may be evaluated using several equivalent approaches.

Peach–Koehler force at the crack tip. For a dislocation representing the crack tip, the Peach–Koehler force per unit length is

$$f_i = \epsilon_{ijk} \sigma_{jl} b_l \xi_k, \quad (86)$$

where ξ is the line direction. Under plane strain conditions, the mode-I stress intensity factor may be obtained as

$$K_I = \sqrt{\frac{2\mu}{1-\nu} f_1^{\text{tip}}}, \quad (87)$$

where f_1^{tip} is the resolved Peach–Koehler force acting on the crack-tip dislocation. The corresponding energy release rate is

$$G = f_1^{\text{tip}}. \quad (88)$$

This relation follows from the equivalence between the work performed by the crack-tip dislocation and the energy released during crack advance.

Contour integral methods. Alternatively, G may be computed using a J -integral evaluated along a contour Γ surrounding the crack tip:

$$J = \int_{\Gamma} \left(W \delta_{1j} - \sigma_{ij} \frac{\partial u_i}{\partial x_1} \right) n_j ds, \quad (89)$$

where W is the strain energy density. In linear elasticity, $J = G$ and is independent of the contour.

Analytical or semi-analytical relations. For benchmark configurations (e.g. straight cracks under remote loading), SIFs may be extracted by fitting the near-tip fields or by using known analytical solutions, providing a useful validation of the numerical procedure.

9.3 Propagation criteria

Once the direction and driving force are known, crack advance is determined by a propagation criterion.

Deterministic threshold criteria. The simplest criterion assumes crack growth occurs when a critical value is exceeded:

$$K_{\text{eq}} \geq K_{IC}, \quad \text{or equivalently} \quad G \geq G_c, \quad (90)$$

where K_{eq} is an appropriate equivalent stress intensity factor (e.g. $\sqrt{K_I^2 + K_{II}^2}$), and K_{IC} or G_c are material fracture toughness parameters.

Rate-dependent and dynamic criteria. For dynamic fracture, propagation may depend on the crack-tip velocity and inertia. In this case, G and K are evaluated using dynamic (retarded) kernels, and crack advance may be governed by a relation of the form

$$\dot{a} = \mathcal{F}(G - G_c), \quad (91)$$

where $\dot{a}$ is the crack growth rate.

Stochastic and microstructure-informed criteria. In materials with distributed microscopic flaws, crack advance may be treated probabilistically. Local fracture resistance is modeled as a random field, and propagation occurs when

$$G(\mathbf{X}_{\text{tip}}) > G_c(\mathbf{X}_{\text{tip}}), \quad (92)$$

with G_c sampled from a prescribed statistical distribution. This approach naturally captures scatter, crack deflection, and intermittent growth.

9.4 Algorithmic Implementation in the DCE Framework

At each load or time increment, the crack growth algorithm proceeds as follows:

1. Evaluate tractions and stresses at each crack tip using the assembled DCE solution.
2. Compute (K_I, K_{II}) and/or G using one of the methods above.
3. Determine the preferred growth direction θ .
4. Check the propagation criterion.
5. If satisfied, insert a new half-crack element at the selected tip with orientation θ , inheriting the appropriate polarity.
6. Retain existing elements; remeshing is performed only if resolution requirements demand it.

This strategy allows for incremental growth, kinking, and branching while preserving the dislocation-based representation of crack-tip singular fields.

10 Scalability and High-Performance Computing

The variational formulation enables:

- Krylov solvers with block preconditioning,
- Fast multipole and hierarchical matrix acceleration,
- GPU/CUDA evaluation of kernel sums,
- Massive parallelism for 10^5 – 10^7 crack elements.

11 Dynamic Fracture and Time-Stepping Formulation

The stress field due to a moving dislocation is written as

$$\sigma_{ij}(\mathbf{X}, t) = \int_{\Gamma} \kappa_{ij,\alpha}^{\text{dyn}}(\mathbf{X} - \mathbf{\Xi}, t - \tau(\mathbf{X}, \mathbf{\Xi})) b_{\alpha}(\mathbf{\Xi}, t) d\Gamma_{\mathbf{\Xi}}, \quad (93)$$

where τ is the retarded time associated with wave propagation at the longitudinal and shear wave speeds.

11.1 Dynamic Stress Kernels

The DCE framework naturally extends to dynamic fracture by replacing static Green's functions with retarded (time-dependent) kernels and evolving the crack geometry in time.

After spatial discretization, the history-dependent traction operator can be written

$$\mathbf{t}(t) = \mathbf{T}_0 \mathbf{q}(t) + \int_0^t \mathbf{T}_{\text{hist}}(t - \tau) \mathbf{q}(\tau) d\tau + \mathbf{t}^\infty(t). \quad (94)$$

With a time step Δt and $t_n = n\Delta t$, a broad class of stable schemes can be written

$$\mathbf{t}^n = \mathbf{T}_0 \mathbf{q}^n + \sum_{k=1}^n \mathbf{T}_k \mathbf{q}^{n-k} + \mathbf{t}_\infty^n. \quad (95)$$

Enforcing traction-free by constrained least squares yields the per-step KKT system

$$\begin{bmatrix} \mathbf{T}_0^T \mathbf{W}^T \mathbf{W} \mathbf{T}_0 & \mathbf{C}^T \\ \mathbf{C} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{q}^n \\ \lambda^n \end{bmatrix} = - \begin{bmatrix} \mathbf{T}_0^T \mathbf{W}^T \mathbf{W} (\sum_{k=1}^n \mathbf{T}_k \mathbf{q}^{n-k} + \mathbf{t}_\infty^n) \\ 0 \end{bmatrix}. \quad (96)$$

11.2 Time discretization

Let t^n denote discrete time levels. Burgers densities and crack geometry are updated incrementally:

$$b^{n+1} = b^n + \Delta t \dot{b}^n, \quad \mathbf{X}_{\text{tip}}^{n+1} = \mathbf{X}_{\text{tip}}^n + \Delta t \mathbf{v}_{\text{tip}}^n. \quad (97)$$

The crack-tip velocity is determined from a kinetic relation, for example

$$\mathbf{v}_{\text{tip}} = v_0 \mathcal{H}(G - G_c) \mathbf{e}_\theta, \quad (98)$$

where $\mathcal{H}$ is a Heaviside function and $\mathbf{e}_\theta$ is the growth direction.

11.3 Dynamic energy balance

At each time step, the dynamic energy release rate satisfies

$$G(t) = f_1^{\text{tip}}(t), \quad (99)$$

where the PK force now includes inertial and wave-mediated contributions. Crack advance occurs only if $G(t) \geq G_c$.

11.4 Worked Example: Kinked Crack under Mixed-Mode Loading

We consider a straight crack of length $2a$ aligned with the global X_1 axis, subjected to a remote stress σ_{22}^∞ . The crack is discretized into four quadratic half-crack elements (two per side).

11.5 Initial solution

Solving the DCE system yields Burgers densities $b_I(x)$ and $b_{II}(x)$ and corresponding stress intensity factors (K_I, K_{II}) at each tip. For symmetric loading,

$$K_{II} = 0, \quad K_I = \sigma_{22}^\infty \sqrt{\pi a}. \quad (100)$$

11.5.1 Determination of kink angle

A small perturbation is applied such that $K_{II} \neq 0$ at the right tip. Using the maximum hoop stress criterion, the kink angle is

$$\theta = 2 \arctan \left[\frac{1}{4} \left(\frac{K_I}{K_{II}} - \sqrt{\left(\frac{K_I}{K_{II}} \right)^2 + 8} \right) \right]. \quad (101)$$

11.5.2 Insertion of a new half-crack

A new half-crack of length Δa is inserted at the right tip with orientation θ and polarity inherited from the parent half-crack. Existing elements are retained without modification.

11.5.3 Updated system and verification

The augmented DCE system is reassembled including interactions between the original and kinked segments. The near-tip fields recover the correct square-root singularity, and the extracted (K_I, K_{II}) at the new tip satisfy the local symmetry condition to leading order.

This example illustrates how crack kinking and directional growth are handled without remeshing the existing crack network, while preserving the physical dislocation-based representation of crack-tip fields.

12 Conclusions and Future Outlook

The variational DCE formulation preserves LEFM-like singular fields via dislocation kernels and singular-weighted basis functions, while enabling scalable solvers through constrained linear algebra. Large-scale crack systems motivate fast operator evaluation (FMM / hierarchical compression), Krylov solvers with preconditioning, and GPU/CUDA implementations of kernel summations and batched linear algebra.

A Sign Conventions and Polarity Algebra

A.1 Kernel invariance

Let $\mathcal{K}(\mathbf{x}, \boldsymbol{\xi}; \mathbf{b}, \mathbf{e}_3)$ denote the stress kernel. Reversing $\mathbf{e}_3$ is equivalent to reversing $\mathbf{b}$:

$$\mathcal{K}(\mathbf{x}, \boldsymbol{\xi}; \mathbf{b}, -\mathbf{e}_3) = \mathcal{K}(\mathbf{x}, \boldsymbol{\xi}; -\mathbf{b}, \mathbf{e}_3).$$

Thus polarity reversal is handled by

$$\mathbf{b}^{(e)} = \hat{\mathbf{b}}^{(e)},$$

with $\hat{\mathbf{b}}^{(e)}$ defined in a right-handed local frame.

A.2 Assembly rule

All influence coefficients take the form

$$A_{ij} = \int_{\Gamma_{e(j)}} \kappa(\mathbf{x}_i, \boldsymbol{\xi}) \pi^{(e(j))} N_j(\boldsymbol{\xi}) d\Gamma_{\boldsymbol{\xi}}.$$

The kernel κ is never modified.

B Relation between Peach–Koehler Force, Energy Release Rate, and Stress Intensity Factors

This appendix establishes the formal connection between the Peach–Koehler (PK) force acting on a crack-tip dislocation, the energy release rate, and the stress intensity factors used in fracture mechanics.

B.1 Peach–Koehler force at the crack tip

For a dislocation with Burgers vector $\mathbf{b}$ and line direction $\boldsymbol{\xi}$, the Peach–Koehler force per unit length is

$$f_i = \epsilon_{ijk} \sigma_{jl} b_l \xi_k. \quad (102)$$

In two-dimensional fracture problems, the dislocation line is parallel to x_3 so that $\boldsymbol{\xi} = \mathbf{e}_3$. The force component relevant to crack advance is f_1 , aligned with the local crack tangent.

B.2 Energy equivalence

Consider a virtual crack extension δa represented by the forward motion of the crack-tip dislocation. The mechanical work performed by the PK force is

$$\delta W = f_1 \delta a. \quad (103)$$

By definition, the energy release rate is

$$G = -\frac{\partial \Pi}{\partial a}, \quad (104)$$

where Π is the total potential energy. Identifying δW with the energy released during crack advance yields

$$\boxed{G = f_1^{\text{tip}}}. \quad (105)$$

B.3 Connection to stress intensity factors

Under linear elastic conditions, the energy release rate is related to the stress intensity factors by

$$G = \begin{cases} \frac{1-\nu^2}{E} (K_I^2 + K_{II}^2), & \text{plane stress,} \\ \frac{1-\nu}{2\mu} (K_I^2 + K_{II}^2), & \text{plane strain.} \end{cases} \quad (106)$$

For pure Mode I fracture in plane strain, this gives

$$\boxed{K_I = \sqrt{\frac{2\mu}{1-\nu} f_1^{\text{tip}}}}. \quad (107)$$

An analogous expression holds for Mode II with the appropriate resolved PK force component.

C VCM_1_0 — Theory (Companion Notes)

2026-01-20

C.1 Dislocation-density representation and jump

For each polyline panel k , the solver returns densities b_I^k , b_{II}^k , local tangent $\mathbf{t}_k$, normal $\mathbf{n}_k$, and panel weight Δs_k .

$$\Delta \mathbf{J}_k = \left(b_{II}^k \mathbf{t}_k + b_I^k \mathbf{n}_k \right) \Delta s_k. \quad (108)$$

The accumulated jump $\mathbf{J}(s)$ yields:

$$\text{COD} = \mathbf{J} \cdot \mathbf{n}, \quad \text{CSD} = \mathbf{J} \cdot \mathbf{t}.$$

C.2 Dipolar (full) vs monopolar (half) constraints

C.3 Dipolar (full) neutrality

$$\int_{\Gamma} b_I(s) ds = 0, \quad \int_{\Gamma} b_{II}(s) ds = 0. \quad (109)$$

C.4 Monopolar (half) junction compatibility and gauge

In the monopolar formulation, branches may carry net Burgers content; compatibility is enforced through junction DOFs and constraint rows in the KKT system.

C.5 Stress and displacement fields (superposition)

Define the panel Burgers increment:

$$\Delta \mathbf{B}_k = \left(b_{II}^k \mathbf{t}_k + b_I^k \mathbf{n}_k \right) \Delta s_k. \quad (110)$$

The global stress field is obtained by superposition:

$$\boldsymbol{\sigma}(\mathbf{x}) = \sum_k \boldsymbol{\sigma}^{\text{edge}}(\mathbf{x} - \mathbf{x}_k, \Delta \mathbf{B}_k) + \boldsymbol{\sigma}^{\infty}. \quad (111)$$

The displacement field is similarly:

$$\mathbf{u}(\mathbf{x}) = \sum_k \mathbf{u}^{\text{edge}}(\mathbf{x} - \mathbf{x}_k, \Delta \mathbf{B}_k) + \mathbf{u}^{\infty}. \quad (112)$$

C.6 KKT constrained least squares structure

$$\begin{bmatrix} K^T K & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} q \\ \lambda \end{bmatrix} = \begin{bmatrix} K^T \text{rhs} \\ 0 \end{bmatrix}. \quad (113)$$

C.7 Peach–Koehler (configurational) force density

$$\mathbf{f}_{PK} = (\boldsymbol{\sigma} \cdot \mathbf{b}) \times \mathbf{e}_z. \quad (114)$$

C.8 Stress intensity factor (SIF) extraction from COD/CSD

Near a crack tip:

$$\text{COD}(r) \approx A_I \sqrt{\frac{r}{2\pi}}, \quad \text{CSD}(r) \approx A_{II} \sqrt{\frac{r}{2\pi}}. \quad (115)$$

Fitting amplitudes gives:

$$K_I = \frac{\mu}{\kappa + 1} A_I, \quad K_{II} = \frac{\mu}{\kappa + 1} A_{II}. \quad (116)$$

with $\kappa = (3 - \nu)/(1 + \nu)$ (plane stress) or $\kappa = 3 - 4\nu$ (plane strain).

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